Kinematics of $\pi^0 \to \gamma\gamma$ Decay

A Pedagogical Solution to Problem E3

Particle Physics — Problem Set Solutions

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1 Physical Setup and Prerequisites

1.1 The Decay $\pi^0 \to \gamma\gamma$

The neutral pion (π^0) is the lightest meson, with mass $m_{\pi^0} \approx 135\ \mathrm{MeV}/c^2$. It decays electromagnetically into two photons with a branching ratio of $\sim 98.8\%$ and a lifetime of $\sim 8.5 \times 10^{-17}$ s.

In this problem we consider a π^0 that is *highly relativistic* in the lab frame, meaning its total energy $E_{\pi^0} \gg m_{\pi^0}c^2$. We define:

$$\gamma = \frac{E_{\pi^0}}{m_{\pi^0}c^2}, \qquad \beta = \sqrt{1 - 1/\gamma^2} \approx 1 - \frac{1}{2\gamma^2}. \tag{1}$$

All π^0's travel in the positive z-direction.

1.2 Center-of-Mass (CM) Frame Recap

In the CM frame (denoted with *), the π^0 is at rest. By energy–momentum conservation:

- Each photon carries energy $E^* = m_{\pi^0}c^2/2$.
- The photons are emitted back-to-back.
- The CM decay angle θ^* is the angle of one photon relative to the boost axis (z).

Key Idea

Because the π^0 has spin-0 and decays into two photons via the electromagnetic interaction, the decay is **isotropic** in the CM frame. That is,

$$\frac{dN}{d(\cos\theta^*)} = \text{const} = \frac{1}{2}\,.$$

(Normalized so $\int_{-1}^{1} \frac{dN}{d(\cos\theta^*)} d(\cos\theta^*) = 1$.)

1.3 Lorentz Boost Along the z-axis

To go from the CM frame to the lab frame, we boost along the z-axis with velocity βc. For a photon with CM energy E^* and CM angle θ^*, the Lorentz transformation gives:

$$E_{\text{lab}} = \gamma\, E^*\big(1 + \beta\cos\theta^*\big), \tag{2}$$

$$\cos\theta_{\text{lab}} = \frac{\cos\theta^* + \beta}{1 + \beta\cos\theta^*}. \tag{3}$$

These are the *only* two equations we need for the entire problem.

2 Part (a): Energy Distribution dn/dE_γ

2.1 Strategy

We want the distribution of photon energies in the lab. We use the standard technique of a **Jacobian change of variables**:

$$\frac{dN}{dE_\gamma} = \frac{dN}{d(\cos\theta^*)} \cdot \left|\frac{d(\cos\theta^*)}{dE_\gamma}\right|.$$

2.2 Derivation

Step 1: Express $\cos\theta^*$ in terms of E_γ.

From Eq. (2):

$$E_\gamma = \gamma E^*(1 + \beta\cos\theta^*)$$

Solving for $\cos\theta^*$:

$$\cos\theta^* = \frac{1}{\beta}\left(\frac{E_\gamma}{\gamma E^*} - 1\right). \tag{4}$$

Step 2: Determine the range of E_γ.

Since $\cos\theta^* \in [-1, +1]$:

$$E_\gamma^{\min} = \gamma E^*(1-\beta), \tag{5}$$
$$E_\gamma^{\max} = \gamma E^*(1+\beta). \tag{6}$$

Now, using $E^* = m_{\pi^0}c^2/2$ and $\gamma = E_{\pi^0}/(m_{\pi^0}c^2)$:

$$\gamma E^* = \frac{E_{\pi^0}}{m_{\pi^0}c^2} \cdot \frac{m_{\pi^0}c^2}{2} = \frac{E_{\pi^0}}{2}.$$

So:

$$E_\gamma^{\min} = \frac{E_{\pi^0}}{2}(1-\beta) \approx \frac{E_{\pi^0}}{4\gamma^2} \approx \frac{m_{\pi^0}^2 c^4}{4E_{\pi^0}} \xrightarrow{\gamma\to\infty} 0, \tag{7}$$
$$E_\gamma^{\max} = \frac{E_{\pi^0}}{2}(1+\beta) \approx E_{\pi^0}. \tag{8}$$

Physics Insight

In the ultra-relativistic limit, the softest photon (emitted backward in the CM) gets $E_\gamma \to 0$, while the hardest photon (emitted forward) takes essentially *all* the pion's energy: $E_\gamma \to E_{\pi^0}$. This enormous asymmetry is a hallmark of highly boosted two-body decays.

Step 3: Compute the Jacobian.
Differentiating Eq. (4):

$$\frac{d(\cos\theta^*)}{dE_\gamma} = \frac{1}{\beta\,\gamma\,E^*}.$$

Step 4: Assemble the distribution.

$$\frac{dN}{dE_\gamma} = \frac{dN}{d(\cos\theta^*)} \cdot \left|\frac{d(\cos\theta^*)}{dE_\gamma}\right| = \frac{1}{2} \cdot \frac{1}{\beta\,\gamma\,E^*}.$$

Since $\beta\,\gamma\,E^* = \beta \cdot E_{\pi^0}/2$:

Part (a) Result

$$\boxed{\frac{dN}{dE_\gamma} = \frac{1}{\beta\,E_{\pi^0}} = \text{const}, \qquad E_\gamma \in \left[E_\gamma^{\min},\ E_\gamma^{\max}\right].} \tag{9}$$

The energy spectrum of decay photons from a highly relativistic π^0 is **flat** (uniform) between $E_\gamma^{\min}$ and $E_\gamma^{\max}$.

2.3 Why Is It Flat? — Physical Intuition

This remarkable result deserves reflection. The isotropic CM distribution $dN/d(\cos\theta^*) = \text{const}$ maps to $dN/dE_\gamma = \text{const}$ because the relationship between E_γ and $\cos\theta^*$ is *linear* (Eq. 2). A linear map preserves uniformity.

In detector physics, this flat spectrum is extremely important: when you reconstruct π^0's from photon pairs, you must cope with one photon often being very soft (low energy) and thus hard to detect.

3 Part (b): Angular Distribution $dn/d\theta$ in the Lab

3.1 Strategy

Now all π^0's travel in the $+z$ direction. We want $dN/d\theta$ (or equivalently $dN/d(\cos\theta)$) in the lab, where θ is the polar angle of a decay photon.

Again, change variables:

$$\frac{dN}{d(\cos\theta)} = \frac{dN}{d(\cos\theta^*)} \cdot \left|\frac{d(\cos\theta^*)}{d(\cos\theta)}\right|.$$

3.2 Derivation

Step 1: Differentiate the boost formula.

From Eq. (3), $\cos\theta = \dfrac{\cos\theta^* + \beta}{1 + \beta\cos\theta^*}$.

We need $d(\cos\theta^*)/d(\cos\theta)$. It is easier to compute $d(\cos\theta)/d(\cos\theta^*)$ and invert.

Let $c^* \equiv \cos\theta^*$, $c \equiv \cos\theta$. Then:

$$c = \frac{c^* + \beta}{1 + \beta\, c^*}.$$

Differentiating (quotient rule):

$$\frac{dc}{dc^*} = \frac{(1 + \beta\, c^*)\cdot 1 - (c^* + \beta)\cdot\beta}{(1 + \beta\, c^*)^2} = \frac{1 + \beta\, c^* - \beta\, c^* - \beta^2}{(1 + \beta\, c^*)^2} = \frac{1 - \beta^2}{(1 + \beta\, c^*)^2}.$$

Since $1 - \beta^2 = 1/\gamma^2$:

$$\frac{d(\cos\theta)}{d(\cos\theta^*)} = \frac{1}{\gamma^2(1 + \beta\cos\theta^*)^2}. \tag{10}$$

Therefore:

$$\frac{d(\cos\theta^*)}{d(\cos\theta)} = \gamma^2(1 + \beta\cos\theta^*)^2. \tag{11}$$

Step 2: Re-express in terms of θ (lab angle).

We need to eliminate $\cos\theta^*$. From the inverse boost:

$$\cos\theta^* = \frac{\cos\theta - \beta}{1 - \beta\cos\theta}. \tag{12}$$

Substituting into $(1 + \beta\cos\theta^*)$:

$$1 + \beta\cos\theta^* = 1 + \beta\cdot\frac{\cos\theta - \beta}{1 - \beta\cos\theta} = \frac{1 - \beta\cos\theta + \beta\cos\theta - \beta^2}{1 - \beta\cos\theta} = \frac{1 - \beta^2}{1 - \beta\cos\theta} = \frac{1}{\gamma^2(1 - \beta\cos\theta)}.$$

Substituting into Eq. (11):

$$\frac{d(\cos\theta^*)}{d(\cos\theta)} = \gamma^2\cdot\frac{1}{\gamma^4(1 - \beta\cos\theta)^2} = \frac{1}{\gamma^2(1 - \beta\cos\theta)^2}.$$

Step 3: Assemble the distribution.

Part (b) Result — $\cos\theta$ distribution

$$\boxed{\frac{dN}{d(\cos\theta)} = \frac{1}{2\gamma^2(1 - \beta\cos\theta)^2}.} \tag{13}$$

If one instead wants $dN/d\theta$:

$$\frac{dN}{d\theta} = \frac{dN}{d(\cos\theta)} \cdot \left|\frac{d(\cos\theta)}{d\theta}\right| = \frac{\sin\theta}{2\gamma^2(1-\beta\cos\theta)^2}.$$

Part (b) Result — θ distribution

$$\boxed{\frac{dN}{d\theta} = \frac{\sin\theta}{2\gamma^2(1-\beta\cos\theta)^2}.} \tag{14}$$

3.3 Physical Features of the Angular Distribution

Physics Insight

The factor $1/(1-\beta\cos\theta)^2$ is the relativistic **beaming** (or headlight) effect. For $\gamma \gg 1$:

- Photons are concentrated in a narrow forward cone of half-angle $\theta \sim 1/\gamma$.
- The distribution is *sharply peaked* at $\theta = 0$.
- Higher E_{π^0} (larger γ) $\Rightarrow$ narrower cone.

This is the same physics that makes synchrotron radiation sharply forward-peaked.

3.4 Maximum Lab Angle

For a photon emitted at $\cos\theta^* = -1$ (directly backward in the CM):

$$\cos\theta = \frac{-1+\beta}{1-\beta} = -1 + \frac{2}{1-\beta} - \frac{2}{1-\beta} + \frac{-1+\beta}{1-\beta}.$$

More directly:

$$\cos\theta_{\text{backward}} = \frac{\beta-1}{1-\beta} = -1.$$

So the backward CM photon still emerges backward in the lab. However, its energy is extremely small: $E = \gamma E^*(1-\beta) \approx m_{\pi^0}^2/(4E_{\pi^0})$.

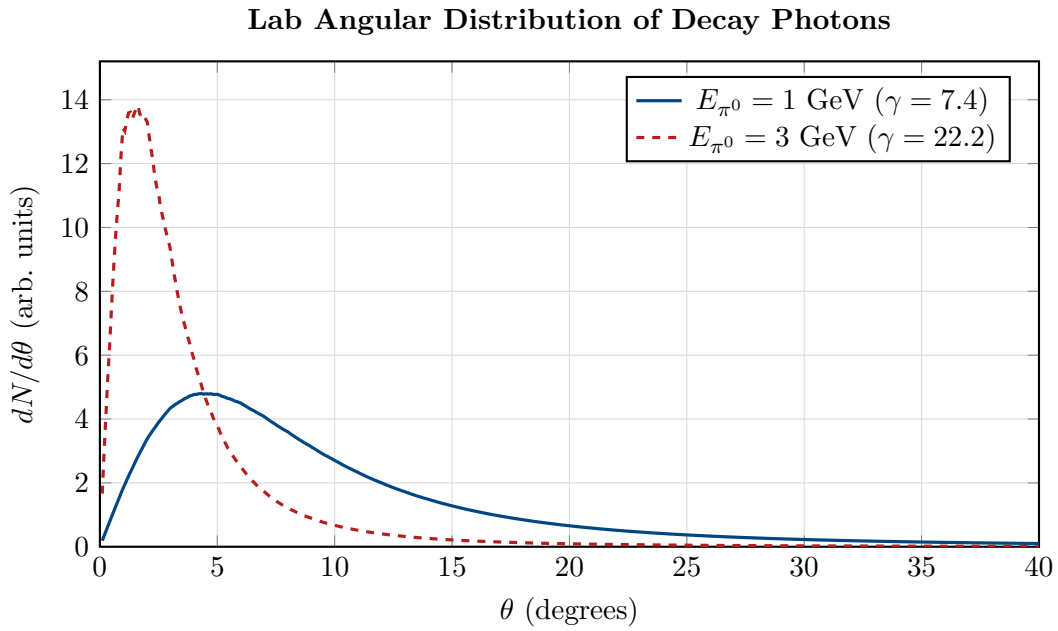
For highly relativistic π^0's, the vast majority of photons are concentrated at small angles $\theta \lesssim 1/\gamma$, but the distribution has a long tail extending to $\theta = \pi$.

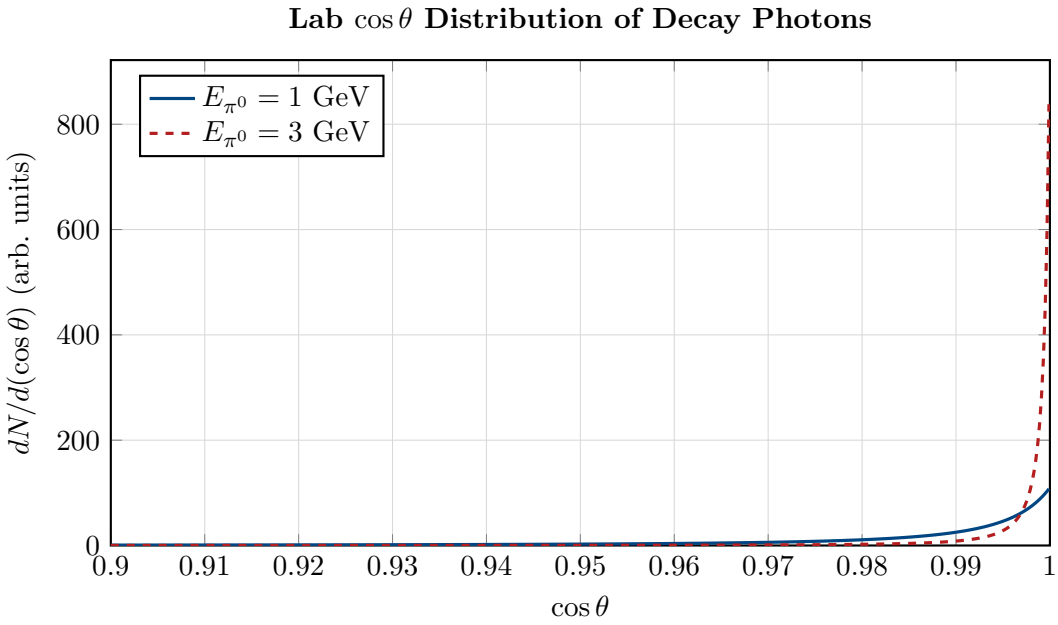
3.5 Plots for 1 GeV and 3 GeV π^0's

For a π^0 of total energy E_{π^0}:

$$\gamma = \frac{E_{\pi^0}}{m_{\pi^0}c^2} = \frac{E_{\pi^0}}{0.135\ \text{GeV}}, \qquad \beta = \sqrt{1-1/\gamma^2}.$$

	$E_{\pi^0} = 1$ GeV	$E_{\pi^0} = 3$ GeV
γ	7.41	22.2
β	0.9909	0.9990
$1/\gamma$ (rad)	0.135	0.045
$1/\gamma$ (deg)	7.7°	2.6°





4 Part (c): Opening Angle Distribution $dn/d(\Delta\theta)$

4.1 What Is the Opening Angle?

The **opening angle** $\Delta\theta \equiv \theta_{12}$ is the angle *between the two photons* in the lab frame. This is **not** the difference $\theta_1 - \theta_2$ of polar angles, but the true 3D angle:

$$\cos(\Delta\theta) = \hat{p}_1 \cdot \hat{p}_2.$$

4.2 Relating $\Delta\theta$ to CM Variables

In the CM frame, the two photons have momenta:

$$\vec{p}_1^{\,*} = E^*(\sin\theta^*,\, 0,\, \cos\theta^*), \tag{15}$$

$$\vec{p}_2^{\,*} = E^*(-\sin\theta^*,\, 0,\, -\cos\theta^*). \tag{16}$$

(They are back-to-back, so photon 2 goes at angle $\pi - \theta^*$.)

After boosting to the lab, each photon's 4-momentum transforms. The lab-frame energies and longitudinal momenta are:

$$E_1 = \gamma E^*(1 + \beta\cos\theta^*), \qquad p_{1z} = \gamma E^*(\cos\theta^* + \beta), \tag{17}$$

$$E_2 = \gamma E^*(1 - \beta\cos\theta^*), \qquad p_{2z} = \gamma E^*(-\cos\theta^* + \beta). \tag{18}$$

The transverse momenta are unchanged:

$$p_{1\perp} = E^*\sin\theta^*, \qquad p_{2\perp} = E^*\sin\theta^*. \tag{19}$$

(Note $p_{2\perp} = +E^*\sin\theta^*$ in magnitude; the sign is opposite in the x-direction.)

The opening angle satisfies:

$$\cos(\Delta\theta) = \frac{\vec{p}_1 \cdot \vec{p}_2}{|\vec{p}_1||\vec{p}_2|} = \frac{p_{1z}p_{2z} - p_{1\perp}^2}{E_1 E_2},$$

where we used $\hat{x}_1 = -\hat{x}_2$ for the transverse components.

Computing the numerator:

$$p_{1z}p_{2z} = \gamma^2(E^*)^2(\cos\theta^* + \beta)(-\cos\theta^* + \beta) = \gamma^2(E^*)^2(\beta^2 - \cos^2\theta^*), \tag{20}$$

$$p_{1\perp}^2 = (E^*)^2\sin^2\theta^* = (E^*)^2(1 - \cos^2\theta^*). \tag{21}$$

So the numerator is:

$$\gamma^2(E^*)^2(\beta^2 - \cos^2\theta^*) - (E^*)^2(1 - \cos^2\theta^*) = (E^*)^2\left[\gamma^2\beta^2 - \gamma^2\cos^2\theta^* - 1 + \cos^2\theta^*\right].$$

Using $\gamma^2\beta^2 = \gamma^2 - 1$:

$$= (E^*)^2\left[\gamma^2 - 1 - 1 + \cos^2\theta^*(1 - \gamma^2)\right] = (E^*)^2\left[\gamma^2 - 2 - (\gamma^2 - 1)\cos^2\theta^*\right].$$

The denominator is:

$$E_1 E_2 = \gamma^2(E^*)^2(1 - \beta^2\cos^2\theta^*).$$

Therefore:

$$\cos(\Delta\theta) = \frac{\gamma^2 - 2 - (\gamma^2 - 1)\cos^2\theta^*}{\gamma^2(1 - \beta^2\cos^2\theta^*)}. \tag{22}$$

4.3 Useful Simplification

This can be rewritten more compactly. Note $\gamma^2 - 1 = \gamma^2\beta^2$:

$$\begin{aligned}\cos(\Delta\theta) &= \frac{\gamma^2(1 - \beta^2\cos^2\theta^*) - 2}{\gamma^2(1 - \beta^2\cos^2\theta^*)} \\ &= 1 - \frac{2}{\gamma^2(1 - \beta^2\cos^2\theta^*)}.\end{aligned} \tag{23}$$

Opening Angle Formula

$$\boxed{\cos(\Delta\theta) = 1 - \frac{2}{\gamma^2(1 - \beta^2\cos^2\theta^*)}.} \tag{24}$$

4.4 Minimum Opening Angle

The minimum opening angle occurs when $\cos^2\theta^* = 0$, i.e., $\theta^* = \pi/2$ (symmetric decay in the CM):

$$\cos(\Delta\theta_{\min}) = 1 - \frac{2}{\gamma^2}.$$

For large γ, using $\cos x \approx 1 - x^2/2$:

$$\Delta\theta_{\min} \approx \frac{2}{\gamma} = \frac{2m_{\pi^0}c^2}{E_{\pi^0}}. \tag{25}$$

Physics Insight

The minimum opening angle is $\Delta\theta_{\min} = 2/\gamma$. This is a famous and practically important result: by measuring the minimum opening angle of photon pairs, you can estimate E_{π^0}. For the maximum opening angle, at $\cos\theta^* = \pm 1$ (asymmetric decay), $\Delta\theta \to \pi$—one photon goes forward, the other backward.

For our numerical values:

	$E_{\pi^0} = 1$ GeV	$E_{\pi^0} = 3$ GeV
γ	7.41	22.2
$\Delta\theta_{\min}$	$2/7.41 = 0.270$ rad $= 15.5°$	$2/22.2 = 0.090$ rad $= 5.2°$

4.5 Deriving $dN/d(\Delta\theta)$

We use the change of variables from $\cos\theta^*$ to $\Delta\theta$. Let $u = \cos^2\theta^*$. From Eq. (24):

$$\cos(\Delta\theta) = 1 - \frac{2}{\gamma^2(1-\beta^2 u)},$$

where $u = \cos^2\theta^*$.

Note that when $\cos\theta^*$ runs from -1 to $+1$, the variable $u = \cos^2\theta^*$ runs from $1 \to 0 \to 1$. There is a **two-fold degeneracy**: both $+\cos\theta^*$ and $-\cos\theta^*$ give the same opening angle. We must account for this factor of 2.

From the opening angle formula, let us solve for u in terms of $\Delta\theta$:

$$1 - \beta^2 u = \frac{2}{\gamma^2(1-\cos\Delta\theta)},$$

$$u = \frac{1}{\beta^2}\left[1 - \frac{2}{\gamma^2(1-\cos\Delta\theta)}\right].$$

Now, $\cos\theta^* = \pm\sqrt{u}$, so $d(\cos\theta^*) = \pm du/(2\sqrt{u})$.

The flat CM distribution gives $dN/d(\cos\theta^*) = 1/2$. Including both signs of $\cos\theta^*$ (the factor of 2 degeneracy):

$$\frac{dN}{d(\Delta\theta)} = 2 \times \frac{1}{2} \times \frac{1}{2\sqrt{u}}\left|\frac{du}{d(\Delta\theta)}\right|.$$

Computing $du/d(\Delta\theta)$:

$$\frac{du}{d(\Delta\theta)} = \frac{1}{\beta^2} \cdot \frac{2\sin(\Delta\theta)}{\gamma^2(1-\cos\Delta\theta)^2}.$$

And we need $\sqrt{u}$:

$$\sqrt{u} = \frac{1}{\beta}\sqrt{1 - \frac{2}{\gamma^2(1-\cos\Delta\theta)}}.$$

Assembling everything:

$$\frac{dN}{d(\Delta\theta)} = \frac{1}{2\sqrt{u}} \cdot \frac{1}{\beta^2} \cdot \frac{2\sin(\Delta\theta)}{\gamma^2(1-\cos\Delta\theta)^2}$$

$$= \frac{\sin(\Delta\theta)}{\beta^2\gamma^2(1-\cos\Delta\theta)^2} \cdot \frac{\beta}{\sqrt{1-\frac{2}{\gamma^2(1-\cos\Delta\theta)}}}.$$

Simplifying:

Part (c) Result

$$\boxed{\frac{dN}{d(\Delta\theta)} = \frac{\sin(\Delta\theta)}{\beta\gamma^2(1-\cos\Delta\theta)^2\sqrt{1-\frac{2}{\gamma^2(1-\cos\Delta\theta)}}}.} \tag{26}$$

Valid for $\Delta\theta \geq \Delta\theta_{\min}$, where $\cos(\Delta\theta_{\min}) = 1 - 2/\gamma^2$.

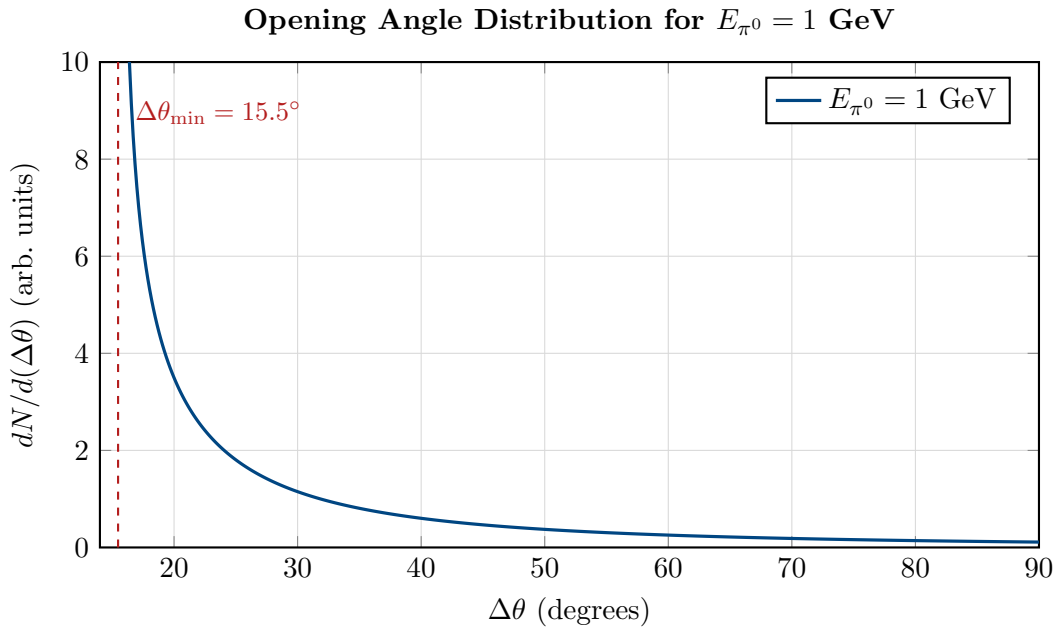
For $\gamma \gg 1$ and small angles, using $1 - \cos\Delta\theta \approx \Delta\theta^2/2$:

Part (c) Small-Angle Approximation

$$\boxed{\frac{dN}{d(\Delta\theta)} \propto \frac{1}{\Delta\theta\sqrt{\Delta\theta^2 - \Delta\theta_{\min}^2}},} \tag{27}$$

where $\Delta\theta_{\min} = 2/\gamma = 2m_{\pi^0}/E_{\pi^0}$. This diverges (integrably) at $\Delta\theta = \Delta\theta_{\min}$.

4.6 Plot for 1 GeV π^0



The distribution shows a sharp peak (integrable singularity) at $\Delta\theta_{\min}$ and falls off at larger angles. Physically, most photon pairs emerge with opening angles close to the minimum value, which corresponds to the symmetric CM decay ($\theta^* = \pi/2$).

5 Summary and Key Takeaways

Quantity	Formula	Shape
dN/dE_γ	$\dfrac{1}{\beta\, E_{\pi^0}}$	Flat between $E_{\min}$ and $E_{\max}$
$dN/d(\cos\theta)$	$\dfrac{1}{2\gamma^2(1-\beta\cos\theta)^2}$	Sharply forward-peaked
$dN/d(\Delta\theta)$	$\propto \dfrac{1}{\Delta\theta\sqrt{\Delta\theta^2-\Delta\theta_{\min}^2}}$	Peaks at $\Delta\theta_{\min} = 2/\gamma$

The Big Picture

All three results flow from two ingredients:

1. **Isotropy in the CM**: $dN/d(\cos\theta^*) = \text{const}$.
2. **The Lorentz boost**: Eqs. (2) and (3).

The flat energy spectrum and forward-peaked angular distribution are universal features of any isotropic two-body decay from a boosted source. The opening-angle distribution's peak at $\Delta\theta_{\min} = 2m/E$ provides a direct mass measurement technique used extensively in calorimeters at collider experiments (e.g., reconstructing π^0's at ALICE, CMS, ATLAS).

A Useful Identities and Checks

Energy–momentum conservation check:

$$E_1 + E_2 = \gamma E^*(1+\beta\cos\theta^*) + \gamma E^*(1-\beta\cos\theta^*) = 2\gamma E^* = E_{\pi^0}. \checkmark$$

Invariant mass of the photon pair:

$$(E_1+E_2)^2 - (\vec{p}_1+\vec{p}_2)^2c^2 = m_{\pi^0}^2c^4. \checkmark$$

This is Lorentz-invariant and can be verified in either frame.

Normalization check for part (a):

$$\int_{E_{\min}}^{E_{\max}} \frac{1}{\beta E_{\pi^0}}\, dE_\gamma = \frac{E_{\max}-E_{\min}}{\beta E_{\pi^0}} = \frac{\gamma E^*(1+\beta)-\gamma E^*(1-\beta)}{\beta E_{\pi^0}} = \frac{2\gamma E^*\beta}{\beta E_{\pi^0}} = \frac{E_{\pi^0}}{E_{\pi^0}} = 1. \checkmark$$